

Coding & Solution Implementation Details

FinRelief AI — Core Logic & Algorithm Documentation

1. Debt Stress & Settlement Calculation Logic (`calculations.py`)

The core mathematical algorithm in backend/utils/calculations.py implements `compute_settlement_metrics()` to calculate Debt-to-Income (DTI), stress score, and recommended settlement amounts:

```
dti = min(100.0, (emi / max(income, 1.0)) * 100.0)
surplus = income - emi - (monthly_expenses if monthly_expenses is not None else income * 0.4)
stress_score = min(100.0, max(0.0, (dti * 0.5) + (min(overdue_days, 180) / 180.0 * 40.0) +
(10.0 if surplus < 0 else 0.0)))
settle_pct = min(70.0, max(20.0, 25.0 + (stress_score * 0.35)))
settle_pct_rounded = round(settle_pct, 1)
settlement_amount = round(amount * settle_pct_rounded / 100.0, 2)
```

2. Gemini AI Integration & Fallback Resilience (`gemini\_client.py`)

The function `generate_letter_body()` invokes Google Gemini (gemini-2.0-flash) using `_generate_with_gemini(prompt)`. The client explicitly catches 8 failure categories:

QUOTA (429), AUTH (401/403), SERVER (5xx), NETWORK (timeout), SAFETY (content block), TRUNCATED (token max), EMPTY (null body), and UNEXPECTED.

Under any failure category, the function transparently returns a legally formatted OTS negotiation template and sets `source = 'fallback'` (or 'gemini' on success).

3. PDF Stream Generation (`pdf\_generator.py`)

`generate_letter_pdf(letter_body, user_name, lender, loan_type)` compiles custom styled PDF documents in memory using ReportLab's SimpleDocTemplate and streams them directly to the client as binary attachments via FastAPI's `StreamingResponse`.

4. Router Module Implementation Overview

- ``routers/auth.py``: Handles `/auth/register` and `/auth/login` with bcrypt hashing and JWT token issuance.
- ``routers/loans.py``: Manages CRUD operations for loans associated with authenticated users.
- ``routers/settlement.py``: Executes settlement calculations and persists `StressSnapshot` records for trend tracking.
- ``routers/letters.py``: Orchestrates letter generation, version history tracking, text editing, and PDF download streaming.
- ``routers/dashboard.py``: Computes aggregate financial KPIs and historical stress trends.